

C Programming - 100 Days Challenge

Days 39 to 49 | Questions 77-98

2D Arrays and Strings

GitHub-ready solutions in simple C with sample input and output.

Day 39

Q77 - Check if the elements on the diagonal of a matrix are distinct.

C Program

```
#include <stdio.h>
int main() {
    int r, c, a[20][20], i, j, distinct = 1;
    scanf("%d %d", &r, &c);
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++) scanf("%d", &a[i][j]);
    int n = r < c ? r : c;
    for (i = 0; i < n; i++)
        for (j = i + 1; j < n; j++)
            if (a[i][i] == a[j][j]) distinct = 0;
    printf(distinct ? "True\n" : "False\n");
    return 0;
}
```

Sample Input

```
3 3
1 2 3
4 5 6
7 8 1
```

Sample Output

```
False
```

Q78 - Find the sum of main diagonal elements for a square matrix.

C Program

```
#include <stdio.h>
int main() {
    int n, a[20][20], i, j, sum = 0;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++) scanf("%d", &a[i][j]);
    for (i = 0; i < n; i++) sum += a[i][i];
    printf("%d\n", sum);
    return 0;
}
```

Sample Input

```
3
1 2 3
4 5 6
7 8 9
```

Sample Output

```
15
```

Day 40

Q79 - Perform diagonal traversal of a matrix.

C Program

```
#include <stdio.h>
int main() {
    int r, c, a[20][20], i, j, s;
    scanf("%d %d", &r, &c);
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++) scanf("%d", &a[i][j]);
    for (s = 0; s <= r + c - 2; s++) {
        for (i = 0; i < r; i++) {
            j = s - i;
            if (j >= 0 && j < c) printf("%d ", a[i][j]);
        }
        return 0;
    }
}
```

Sample Input

```
3 3
1 2 3
4 5 6
7 8 9
```

Sample Output

```
1 2 4 3 5 7 6 8 9
```

Q80 - Multiply two matrices.

C Program

```
#include <stdio.h>
int main() {
    int a[10][10], b[10][10], p[10][10] = {0};
    int r1, c1, r2, c2, i, j, k;
    scanf("%d %d", &r1, &c1);
    for(i=0; i<r1; i++) for(j=0; j<c1; j++) scanf("%d", &a[i][j]);
    scanf("%d %d", &r2, &c2);
    for(i=0; i<r2; i++) for(j=0; j<c2; j++) scanf("%d", &b[i][j]);
    if(c1 != r2) { printf("Multiplication not possible\n"); return 0; }
    for(i=0; i<r1; i++)
        for(j=0; j<c2; j++)
            for(k=0; k<c1; k++) p[i][j] += a[i][k]*b[k][j];
    for(i=0; i<r1; i++) {
        for(j=0; j<c2; j++) printf("%d ", p[i][j]);
        printf("\n");
    }
    return 0;
}
```

Sample Input

```
2 2
1 2
3 4
2 2
5 6
7 8
```

Sample Output

```
19 22
43 50
```

Day 41

Q81 - Count characters in a string without using built-in length functions.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int count = 0;
    fgets(s, sizeof(s), stdin);
    while(s[count] != '\0' && s[count] != '\n') count++;
    printf("%d\n", count);
    return 0;
}
```

Sample Input

Hello

Sample Output

5

Q82 - Print each character of a string on a new line.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int i = 0;
    fgets(s, sizeof(s), stdin);
    while(s[i] != '\0' && s[i] != '\n') {
        printf("%c\n", s[i]); i++;
    }
    return 0;
}
```

Sample Input

Code

Sample Output

C
o
d
e

Day 42

Q83 - Count vowels and consonants in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200], ch; int i=0, v=0, c=0;
    fgets(s, sizeof(s), stdin);
    while(s[i] != '\0') {
        ch=s[i];
        if(ch>='A'&&ch<='Z') ch=ch+32;
        if(ch>='a'&&ch<='z') {
            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u') v++;
            else c++;
        }
        i++;
    }
    printf("Vowels = %d\nConsonants = %d\n", v, c);
    return 0;
}
```

Sample Input

Hello World

Sample Output

Vowels = 3
Consonants = 7

Q84 - Convert a lowercase string to uppercase without using built-in functions.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int i=0;
    fgets(s, sizeof(s), stdin);
    while(s[i] != '\0') {
        if(s[i]>='a' && s[i]<='z') s[i]=s[i]-32;
        i++;
    }
    printf("%s", s);
    return 0;
}
```

Sample Input

hello world

Sample Output

HELLO WORLD

Day 43

Q85 - Reverse a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int n=0, i;
    fgets(s, sizeof(s), stdin);
    while(s[n]!='\0' && s[n]!='\n') n++;
    for(i=n-1;i>=0;i--) printf("%c",s[i]);
    printf("\n");
    return 0;
}
```

Sample Input

hello

Sample Output

olleh

Q86 - Check if a string is a palindrome.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int n=0,i,ok=1;
    fgets(s,sizeof(s),stdin);
    while(s[n]!='\0' && s[n]!='\n') n++;
    for(i=0;i<n/2;i++) if(s[i]!=s[n-1-i]) ok=0;
    printf(ok ? "Palindrome\n" : "Not palindrome\n");
    return 0;
}
```

Sample Input

madam

Sample Output

Palindrome

Day 44

Q87 - Count spaces, digits, and special characters in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200], ch; int i=0, sp=0, d=0, special=0;
    fgets(s, sizeof(s), stdin);
    while(s[i]!='\0' && s[i]!='\n') {
        ch=s[i];
        if(ch==' ') sp++;
        else if(ch>='0'&&ch<='9') d++;
        else if(!((ch>='a'&&ch<='z')||(ch>='A'&&ch<='Z')) special++;
        i++;
    }
    printf("Spaces = %d\nDigits = %d\nSpecial = %d\n", sp, d, special);
    return 0;
}
```

Sample Input

Hello 123!

Sample Output

Spaces = 1
Digits = 3
Special = 1

Q88 - Replace spaces with hyphens in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int i=0;
    fgets(s, sizeof(s), stdin);
    while(s[i]!='\0') { if(s[i]==' ') s[i]='-'; i++; }
    printf("%s", s);
    return 0;
}
```

Sample Input

C programming is easy

Sample Output

C-programming-is-easy

Day 45

Q89 - Count frequency of a given character in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200], ch; int i=0, count=0;
    fgets(s, sizeof(s), stdin);
    scanf("%c", &ch);
    while(s[i]!='\0' && s[i]!='\n') { if(s[i]==ch) count++; i++; }
    printf("%d\n", count);
    return 0;
}
```

Sample Input

banana
a

Sample Output

3

Q90 - Toggle case of each character in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int i=0;
    fgets(s, sizeof(s), stdin);
    while(s[i]!='\0') {
        if(s[i]>='a'&&s[i]<='z') s[i]-=32;
        else if(s[i]>='A'&&s[i]<='Z') s[i]+=32;
        i++;
    }
    printf("%s", s);
    return 0;
}
```

Sample Input

HeLLo

Sample Output

hELlO

Day 46

Q91 - Remove all vowels from a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200], ch; int i=0;
    fgets(s,sizeof(s),stdin);
    while(s[i]!='\0' && s[i]!='\n') {
        ch=s[i];
        if(!(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
            ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')) printf("%c",ch);
        i++;
    }
    printf("\n"); return 0;
}
```

Sample Input

education

Sample Output

dctn

Q92 - Find the first repeating lowercase alphabet in a string.

C Program

```
#include <stdio.h>
int main() {
    char s[200]; int seen[26]={0}, i=0;
    fgets(s,sizeof(s),stdin);
    while(s[i]!='\0' && s[i]!='\n') {
        if(s[i]>='a' && s[i]<='z') {
            int x=s[i]-'a';
            if(seen[x]) { printf("%c\n",s[i]); return 0; }
            seen[x]=1;
        }
        i++;
    }
    printf("No repeating alphabet\n");
    return 0;
}
```

Sample Input

programming

Sample Output

r

Day 47

Q93 - Check if two strings are anagrams of each other.

C Program

```
#include <stdio.h>
int main() {
    char a[200], b[200]; int f[256]={0}, i=0;
    fgets(a, sizeof(a), stdin); fgets(b, sizeof(b), stdin);
    while(a[i]!='\0' && a[i]!='\n'){ f[(unsigned char)a[i]]++; i++; }
    i=0;
    while(b[i]!='\0' && b[i]!='\n'){ f[(unsigned char)b[i]]--; i++; }
    for(i=0; i<256; i++) if(f[i]!=0){ printf("Not anagrams\n"); return 0; }
    printf("Anagrams\n"); return 0;
}
```

Sample Input

listen
silent

Sample Output

Anagrams

Q94 - Find the longest word in a sentence.

C Program

```
#include <stdio.h>
int main() {
    char s[300], word[100], longest[100];
    int i=0, j=0, k, max=0;
    fgets(s, sizeof(s), stdin);
    while(1) {
        if(s[i]!=' ' && s[i]!='\n' && s[i]!='\0') word[j++]=s[i];
        else {
            if(j>max){ max=j; for(k=0; k<j; k++) longest[k]=word[k]; longest[j]='\0'; }
            j=0;
            if(s[i]=='\n' || s[i]=='\0') break;
        }
        i++;
    }
    printf("%s\n", longest); return 0;
}
```

Sample Input

C programming is interesting

Sample Output

programming

Day 48

Q95 - Check if one string is a rotation of another.

C Program

```
#include <stdio.h>
#include <string.h>
int main() {
    char a[100], b[100], temp[200];
    scanf("%99s", a); scanf("%99s", b);
    if(strlen(a) != strlen(b)) { printf("Not rotation\n"); return 0; }
    strcpy(temp, a); strcat(temp, a);
    printf(strstr(temp, b) ? "Rotation\n" : "Not rotation\n");
    return 0;
}
```

Sample Input

```
abcde
deabc
```

Sample Output

```
Rotation
```

Q96 - Reverse each word in a sentence without changing the word order.

C Program

```
#include <stdio.h>
int main() {
    char s[300]; int i=0, start, j;
    fgets(s, sizeof(s), stdin);
    while(s[i] != '\0') {
        while(s[i] == ' ') { printf(" "); i++; }
        start = i;
        while(s[i] != ' ' && s[i] != '\n' && s[i] != '\0') i++;
        for(j = i-1; j >= start; j--) printf("%c", s[j]);
        if(s[i] == '\n' || s[i] == '\0') break;
    }
    printf("\n"); return 0;
}
```

Sample Input

```
Hello World from C
```

Sample Output

```
olleH dlrow morf C
```

Day 49

Q97 - Print the initials of a name.

C Program

```
#include <stdio.h>
int main() {
    char name[200]; int i=0;
    fgets(name, sizeof(name), stdin);
    if(name[0]!=' ' && name[0]!='\n') printf("%c", name[0]);
    while(name[i]!='\0') {
        if(name[i]==' ' && name[i+1]!=' ' && name[i+1]!='\n' && name[i+1]!='\0')
            printf("%c", name[i+1]);
        i++;
    }
    printf("\n"); return 0;
}
```

Sample Input

Ansh Kumar Rana

Sample Output

A.K.R.

Q98 - Print initials of a name with the surname displayed in full.

C Program

```
#include <stdio.h>
#include <string.h>
int main() {
    char name[200]; int i, last=0;
    fgets(name, sizeof(name), stdin);
    name[strcspn(name, "\n")]='\0';
    for(i=0; name[i]!='\0'; i++) if(name[i]==' ') last=i+1;
    if(last==0) { printf("%s\n", name); return 0; }
    printf("%c. ", name[0]);
    for(i=1; i<last-1; i++)
        if(name[i]==' ' && i+1<last) printf("%c. ", name[i+1]);
    printf("%s\n", name+last);
    return 0;
}
```

Sample Input

Ansh Kumar Rana

Sample Output

A. K. Rana